

The original zeta endpoint, the heat range, and the arithmetic prime class

Split-Zero research programme

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The positive-time theta filter has no tempered preimage of the actual theta heat function. Its missing range component is exactly the class of the endpoint constant $1/16$, fixed by the residue of the original zeta function. This note proves the full Schwartz range and inverse, constructs a faithful source retaining the two endpoint continuations, and evaluates the resulting prime-boundary class. It then gives an invertible source filter and the complete original-zeta arithmetic pairing for that test. No sign of that final pairing is assumed.

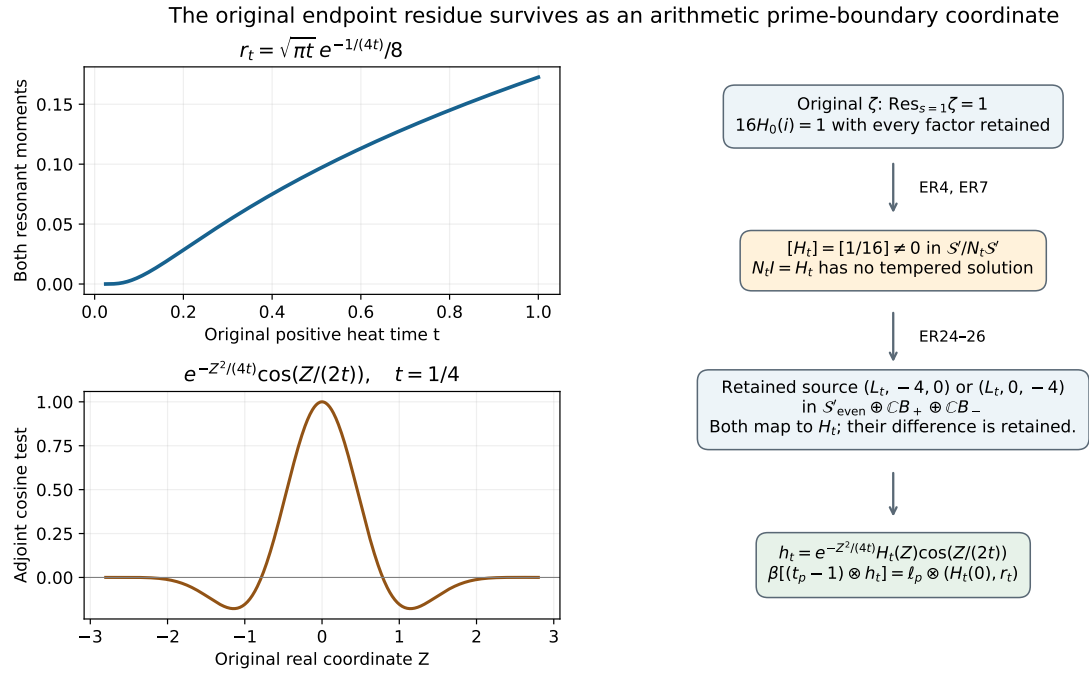


Figure 1: Exact moments ER7 and an adjoint test at the stated original time, with the maps proved in ER21, ER24–30. The last arrow constructs the displayed Schwartz test from H_t ; it does not identify a distribution source with its arithmetic quotient. The human theta source is Rodgers–Tao, equations `hoz`, `sas`, `phidef`, `htdef`; the arithmetic periodization source is Connes, Section III (6)–(19), as cited in the proof. The plotted curves are the exact displayed functions, not computed zero data.

1 The missing heat endpoint is detected by the original arithmetic residue

We calculate a range obstruction, its exact inverse on the range, and a faithful enlarged source. The obstruction has an evaluated value fixed by the residue of the original Riemann zeta function.

It also determines a nonzero class in the existing prime-boundary quotient. All calculations use the original coordinate $s = \frac{1}{2} + iZ/2$, the original time t , and the complete multiplier

$$C(s) = s(s-1)\pi^{-s/2}\Gamma(s/2). \quad (\text{ER1})$$

The relevant source is Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, original author TeX, equations `hoz`, `sas`, `phidef`, and `htdef`. We use its exact theta density and prove the new range and receiving statements below. No assumption about the location of nontrivial zeros enters this calculation.

1.1 The exact resonant moments

Write

$$\Phi(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \quad H_t(Z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du. \quad (\text{ER2})$$

Poisson summation gives the even extension of Φ . Every derivative of this extension decreases faster than any Gaussian at both ends. One can verify the latter assertion directly on $u \geq 0$ by termwise differentiation of ER2: every differentiated term is a polynomial in $n^2 e^{4u}$ times fixed exponentials in u and $e^{-\pi n^2 e^{4u}}$; its sum has the asserted bound. Evenness gives the assertion on $u \leq 0$. Thus H_t restricted to the real Z -axis is Schwartz for every real t , by repeated integration by parts in the whole-line Fourier integral of the even density. It is also entire in Z .

The original theta identity is

$$16H_0(-2i(s - \tfrac{1}{2})) = C(s)\zeta(s). \quad (\text{ER3})$$

Here every product at an exceptional point means its meromorphic continuation with both factors retained. In particular

$$\lim_{s \rightarrow 1} (s-1)\zeta(s) = 1, \quad \lim_{s \rightarrow 1} \frac{C(s)}{s-1} = 1 \cdot \pi^{-1/2}\Gamma(1/2) = 1, \quad H_0(i) = \frac{1}{16}. \quad (\text{ER4})$$

These limits also follow from the full theta formula

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = -\frac{1}{s} + \frac{1}{s-1} + \int_1^\infty \psi(x) \{x^{s/2-1} + x^{(1-s)/2-1}\} dx, \quad \psi(x) = \sum_{n \geq 1} e^{-\pi n^2 x}. \quad (\text{ER5})$$

The integral is entire, with locally uniform derivative convergence. Consequently the residue in ER4 is an original endpoint contribution, not a substituted hypothesis about a completed function.

For $t > 0$, put

$$\omega_t = \frac{1}{2t}, \quad \ell_{\pm, t}(h) = \int_{\mathbb{R}} e^{-Z^2/(4t)} e^{\pm iZ/(2t)} h(Z) dZ. \quad (\text{ER6})$$

For tempered distributions these integrals mean the complex-bilinear distribution pairing with the displayed Schwartz test, with no conjugation. Gaussian integration gives

$$\boxed{\ell_{+, t}(H_t) = \ell_{-, t}(H_t) = 2\sqrt{\pi t} e^{-1/(4t)} H_0(i) = \frac{\sqrt{\pi t}}{8} e^{-1/(4t)} =: r_t > 0.} \quad (\text{ER7})$$

Indeed the inner Gaussian integral against $\cos(Zu)$ is

$$\sqrt{\pi t} \{e^{-t(u+\omega_t)^2} + e^{-t(u-\omega_t)^2}\} = 2\sqrt{\pi t} e^{-tu^2-1/(4t)} \cosh u. \quad (\text{ER8})$$

Fubini is justified by the finite integral of $e^{tu^2} |\Phi(u)|$ times the absolute Gaussian integral. Multiplication by the retained e^{tu^2} in ER2 yields ER7. Thus neither the heat time nor the endpoint factor has been dropped.

1.2 An exact range theorem on the original real axis

Keep the full differential filter

$$N_t = -\frac{1 + (Z - 2t\partial_Z)^2}{64} = -\frac{1 + Z^2 - 2t - 4tZ\partial_Z + 4t^2\partial_Z^2}{64}. \quad (\text{ER9})$$

It acts continuously on both $\mathcal{S}(\mathbb{R})$ and $\mathcal{S}'(\mathbb{R})$. For the bilinear distribution pairing its transpose is

$$N_t^\top = -\frac{1 + (Z + 2t\partial_Z)^2}{64}, \quad N_t^\top(e^{-Z^2/(4t)}e^{\pm iZ/(2t)}) = 0. \quad (\text{ER10})$$

The last equality follows already from $(Z + 2t\partial_Z)e^{-Z^2/(4t)}e^{\pm iZ/(2t)} = \pm ie^{-Z^2/(4t)}e^{\pm iZ/(2t)}$. Therefore $\ell_{\pm,t}(N_t I) = 0$ for every tempered I . ER7 proves

$$\boxed{\text{There is no } I \in \mathcal{S}'(\mathbb{R}) \text{ with } N_t I = H_t, \quad t > 0.} \quad (\text{ER11})$$

This is a statement about a specified map and domain.

The precise Schwartz range is

$$0 \longrightarrow \mathcal{S}(\mathbb{R}) \xrightarrow{N_t} \mathcal{S}(\mathbb{R}) \xrightarrow{(\ell_{+,t}, \ell_{-,t})} \mathbb{C}^2 \longrightarrow 0. \quad (\text{ER12})$$

We prove exactness and give an inverse on the first image. For $h \in \mathcal{S}$ with both moments zero, set

$$(\mathcal{G}_t h)(Z) = -\frac{32}{t} e^{Z^2/(4t)} \int_{-\infty}^Z \sin \frac{Z-y}{2t} e^{-y^2/(4t)} h(y) dy. \quad (\text{ER13})$$

Twice differentiating the absolutely convergent integral proves $N_t \mathcal{G}_t h = h$. For clarity, after the exact substitution $g = e^{-Z^2/(4t)} I$, the equation is

$$g'' + \frac{g}{4t^2} = -\frac{16}{t^2} e^{-Z^2/(4t)} h. \quad (\text{ER14})$$

The coefficient in ER13 is the coefficient in ER14 divided by ω_t . Since both moments vanish, the full-line sine integral vanishes for each Z . On $Z > 0$ ER13 can therefore be written using minus the complementary integral from Z to $+\infty$. For either sign of Z , with $|Z| \geq 1$, this gives

$$(\mathcal{G}_t h)(Z) = -\frac{32}{t} \int_0^\infty \sin \frac{v}{2t} e^{-|Z|v/(2t) - v^2/(4t)} h(Z + \text{sgn}(Z)v) dv. \quad (\text{ER15})$$

Every derivative is bounded by a finite sum of the derivatives of h times powers of v , with the same exponential majorant. For every integer $j \geq 0$, $\int_0^\infty v^j e^{-|Z|v/(2t) - v^2/(4t)} dv = j!(2t/|Z|)^{j+1}$. Together with $1 + |Z + \text{sgn}(Z)v| \geq 1 + |Z|$, this proves all Schwartz estimates, with each output seminorm bounded by finitely many input seminorms. The integral on a bounded interval has the same property by Gaussian domination. Hence $\mathcal{G}_t : \ker(\ell_+, \ell_-) \rightarrow \mathcal{S}$ is continuous.

The homogeneous equation ER14 has the complete distribution solution $g = c_+ e^{iZ/(2t)} + c_- e^{-iZ/(2t)}$. This can be proved by factoring the constant-coefficient differential operator into its two distinct first-order factors and solving each distribution equation by an integrating factor. Consequently every homogeneous I is

$$I = e^{Z^2/(4t)} (c_+ e^{iZ/(2t)} + c_- e^{-iZ/(2t)}). \quad (\text{ER16})$$

No nonzero such function defines a tempered distribution. To prove this without an unsupported growth criterion, the parenthesized function $p(Z)$ is periodic and nonzero on an interval. Choose a nonnegative compact smooth function χ on such an interval. Translate $\chi \bar{p}$ by integer multiples of the period $4\pi t$. The pairing with ER16 is the integral of $e^{Z^2/(4t)} \chi |p|^2$, which grows faster than any polynomial in the translation parameter, whereas every Schwartz seminorm of these

translated tests grows at most polynomially. This contradicts tempered continuity. Thus N_t is injective even on $\mathcal{S}'$.

Finally, the moment map is onto. For

$$\varphi_t(Z) = e^{-Z^2/(4t)}, \quad A_t = \sqrt{2\pi t} e^{-1/(8t)}$$

direct Gaussian differentiation gives

$$\ell_{\pm,t}(a\varphi_t + bZ\varphi_t) = A_t(a \pm ib/2). \quad (\text{ER17})$$

For a requested pair u_+, u_- , take $a = (u_+ + u_-)/(2A_t)$ and $b = (u_+ - u_-)/(iA_t)$. This proves ER12, including a specified continuous right inverse. On the original even source the exact sequence instead has $\mathcal{S}_{\text{even}}$, $\mathcal{S}_{\text{even}}$, and $\mathbb{C}$, with its last map $\ell_{+,t} = \ell_{-,t}$.

In particular, without suppressing the complementary term,

$$H_t = N_t I_t^{\text{Sch}} + \frac{e^{-1/(8t)}}{8\sqrt{2}} \varphi_t, \quad I_t^{\text{Sch}} = \mathcal{G}_t \left(H_t - \frac{e^{-1/(8t)}}{8\sqrt{2}} \varphi_t \right). \quad (\text{ER18})$$

This is the explicitly chosen Gaussian section of ER12; it is not declared to be a unique choice of section.

1.3 The original endpoint gives a faithful enlargement

The actual theta density before filtering is $\rho(u) = 8e^u \psi(e^{4u}) - 4e^{-u}$. Put

$$L_t(Z) = 8 \int_0^\infty e^{tu^2+u} \psi(e^{4u}) \cos(Zu) du. \quad (\text{ER19})$$

Every derivative of L_t is bounded on the real axis, by absolute integration of its differentiated density. It defines a tempered distribution. Integration by parts with all zero-endpoint terms gives

$$N_t L_t = H_t - \frac{1}{16}. \quad (\text{ER20})$$

For completeness the theta-only density in ER19 has derivative -4 at zero: differentiating Poisson reflection gives $\psi'(1) = -\psi(1)/4 - 1/8$. For any suitable density r , the full calculation is

$$\int_0^\infty e^{tu^2} (r'' - r) \cos(Zu) du = -r'(0) - [1 + (Z - 2t\partial_Z)^2] \int_0^\infty e^{tu^2} r \cos(Zu) du.$$

The interior density for ER19 is 64Φ . Substitution yields ER20, including its sign. Thus the exact nonzero quotient class is

$$[H_t] = [1/16] \neq 0 \quad \text{in } \mathcal{S}'/N_t \mathcal{S}'. \quad (\text{ER21})$$

No general dimension assertion about this distribution quotient is required here. The equality and its nonvanishing are fully evaluated.

Here are the actual two endpoint continuations. For $q > 0$ set

$$F(q, a) = \frac{\sqrt{\pi}}{2\sqrt{q}} e^{a^2/(4q)} \operatorname{erfc} \frac{a}{2\sqrt{q}}, \quad B(q, Z) = \frac{F(q, 1 - iZ) + F(q, 1 + iZ)}{2}. \quad (\text{ER22})$$

They equal their Gaussian integrals by completing the square. Use the square root positive for positive q , and approach $q = -t$ from its upper or lower half-plane to obtain B_+ and B_- . The entire identity $\operatorname{erfc}(z) + \operatorname{erfc}(-z) = 2$ gives

$$B_+ - B_- = -i\sqrt{\frac{\pi}{t}} e^{(Z^2-1)/(4t)} \cos \frac{Z}{2t}, \quad N_t B_\pm = -\frac{1}{64}. \quad (\text{ER23})$$

The second equation first follows for $q = -t > 0$ from the retained endpoint derivative $(e^{-u})'(0) = -1$; analytic continuation gives both lateral values. This is the erfc convention of N. M. Temme, [DLMF 7.2.2](#).

Define a precise enlarged even source, as a subspace of distributions, by the map

$$\iota_t : \mathcal{S}'_{\text{even}} \oplus \mathbb{C}^2 \longrightarrow \mathcal{D}'_{\text{even}}, \quad (U, c_+, c_-) \longmapsto U + c_+ B_+ + c_- B_-. \quad (\text{ER24})$$

This map is injective. Indeed, if the last two terms are tempered, apply N_t and ER10. The nonzero moments of the constant show $c_+ + c_- = 0$. ER23 and the translated-test proof following ER16 then give $c_+ = c_- = 0$. Its filter is exactly

$$N_t \iota_t(U, c_+, c_-) = N_t U - \frac{c_+ + c_-}{64}. \quad (\text{ER25})$$

Its kernel has precisely the one-dimensional coordinate line $(0, c, -c)$: necessity follows again from ER10 and the injectivity of N_t on $\mathcal{S}'$; sufficiency follows from ER23. The two original unfiltered continuations are the distinct vectors

$$I_+ = \iota_t(L_t, -4, 0), \quad I_- = \iota_t(L_t, 0, -4), \quad N_t I_{\pm} = H_t. \quad (\text{ER26})$$

They satisfy the heat equation on their actual sectorial domains by differentiating the convergent integral for negative time and continuing analytically. The enlarged source has retained, rather than identified, both endpoint values. The full original-zeta returns are $I_{\pm}/\{\pi^{-s/2}\Gamma(s/2)\}$ and $16H_t/C(s)$, at $Z = -2i(s - \frac{1}{2})$. Their exact map is the filter ER9 with these displayed factors, including its derivative terms; no equality between the two returned families is asserted.

The boundary values remember the chosen continuation. For an actual solution I of $N_t I = H_t$, let $g = e^{-Z^2/(4t)} I$. For $\varepsilon = \pm 1$ the complete Green boundary is

$$\begin{aligned} W_{\varepsilon}[I](Z) &= e^{\varepsilon i Z/(2t)} [g'(Z) - \varepsilon i g(Z)/(2t)] \\ &= e^{-Z^2/(4t)} e^{\varepsilon i Z/(2t)} \left[I'(Z) - \left(\frac{Z}{2t} + \frac{\varepsilon i}{2t} \right) I(Z) \right], \quad (\text{ER26a}) \\ W_{\varepsilon}[I]'(Z) &= -\frac{16}{t^2} e^{-Z^2/(4t)} e^{\varepsilon i Z/(2t)} H_t(Z). \end{aligned}$$

The last identity follows from ER14 by differentiation. Equivalently, integration of the full original differential operator over $[a, b]$ gives

$$\int_a^b (\psi_{\varepsilon} N_t I - I N_t^{\top} \psi_{\varepsilon}) dZ = -\frac{t^2}{16} [W_{\varepsilon}[I]]_a^b, \quad \psi_{\varepsilon} = e^{-Z^2/(4t)} e^{\varepsilon i Z/(2t)}. \quad (\text{ER26b})$$

The drift term contributes $-Z\psi_{\varepsilon}I/t$ inside the second-derivative Green bracket $\psi_{\varepsilon}I' - \psi'_{\varepsilon}I - Z\psi_{\varepsilon}I/t$. Inserting ψ'_{ε} gives exactly ER26b. The right side of ER26a is integrable on the real line. Both Wronskian limits exist, and their difference is

$$W_{\varepsilon}(+\infty) - W_{\varepsilon}(-\infty) = -2\sqrt{\pi} t^{-3/2} e^{-1/(4t)}. \quad (\text{ER26c})$$

The separate limits can also be calculated. Write $c_t = \sqrt{\pi/t} e^{-1/(4t)}$. Direct substitution in ER22 on the upper q -branch gives

$$\begin{aligned} e^{-Z^2/(4t)} I_+ &= e^{-Z^2/(4t)} L_t \\ &+ i c_t \left\{ e^{iZ/(2t)} \operatorname{erfc} \frac{-Z-i}{2\sqrt{t}} + e^{-iZ/(2t)} \operatorname{erfc} \frac{Z-i}{2\sqrt{t}} \right\}. \quad (\text{ER26d}) \end{aligned}$$

For $x > 0$ and fixed real y ,

$$\operatorname{erfc}(x + iy) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-(v+iy)^2} dv, \quad |\operatorname{erfc}(x + iy)| \leq \frac{e^{y^2-x^2}}{\sqrt{\pi}x}.$$

The first formula follows by moving the entire primitive to a horizontal ray; the extra vertical segment tends to zero at infinity. The bound follows by comparison with the real Gaussian tail. Its derivative has the Gaussian bound from $\operatorname{erfc}'(z) = -2e^{-z^2}/\sqrt{\pi}$. Use also $\operatorname{erfc}(-z) = 2 - \operatorname{erfc}(z)$. Since L_t, L'_t are bounded, ER26d and ER23 now show that $e^{-Z^2/(4t)}I_+$ tends to $2ic_te^{iZ/(2t)}$ at $+\infty$ and $2ic_te^{-iZ/(2t)}$ at $-\infty$, with remainders and their derivatives tending to zero. For I_- the corresponding terms are $-2ic_te^{-iZ/(2t)}$ and $-2ic_te^{iZ/(2t)}$. Substitution into ER26a proves

	$W_+(-\infty)$	$W_+(+\infty)$	$W_-(-\infty)$	$W_-(+\infty)$	
I_+	$2c_t/t$	0	0	$-2c_t/t$	(ER26e)
I_-	0	$-2c_t/t$	$2c_t/t$	0.	

Every row retains the same compulsory difference ER26c.

On the actual arithmetic line put $A(s) = \pi^{-s/2}\Gamma(s/2)$ and $f_\pm(t, s) = I_\pm(t, -2i(s - \frac{1}{2}))/A(s)$. There A is nonzero and holomorphic. Since $\partial_Z = (i/2)\partial_s$, the same boundary becomes

$$\begin{aligned} W_\varepsilon[I_\pm] &= \exp \frac{(s - \frac{1}{2})^2 + \varepsilon(s - \frac{1}{2})}{t} \\ &\quad \times \left\{ \frac{i}{2}[A'f_\pm + A\partial_s f_\pm] + \frac{i}{t} \left(s - \frac{1+\varepsilon}{2} \right) A f_\pm \right\}, \\ A'(s) &= A(s) \left[-\frac{\log \pi}{2} + \frac{1}{2}\psi_\Gamma(s/2) \right]. \end{aligned} \tag{ER26f}$$

Both multiplier derivatives and both endpoint limits are retained. This gives the exact original-zeta boundary receiving map, including the data changed by choosing a different lateral continuation.

1.4 An evaluated prime-boundary class and its full zeta transform

There is an arithmetic receiver for the nonzero obstruction value. Define the actual even Schwartz source

$$h_t(Z) = e^{-Z^2/(4t)} H_t(Z) \cos \frac{Z}{2t}. \tag{ER27}$$

Its original two moments are

$$\left(h_t(0), \int_{\mathbb{R}} h_t(Z) dZ \right) = (H_t(0), r_t). \tag{ER28}$$

Let $H = \mathcal{S}_{\text{even}}(\mathbb{R})$, $H_0 = \{h : h(0) = \int h = 0\}$, $R = \mathbb{C}[\mathbb{Q}_{>0}^\times]$, and $I = \ker(\text{aug} : R \rightarrow \mathbb{C})$. For $M_0 = \ker(\text{aug} \otimes (h \mapsto (h(0), \int h)))$, the existing quotient is $Q_0 = M_0/IM_0$. Its exact prime-boundary coordinate is

$$\beta \left[\sum_a t_a \otimes h_a \right] = \sum_p \ell_p \otimes \sum_a v_p(a) \left(h_a(0), \int h_a \right). \tag{ER29}$$

To recall why this is a well-defined detecting map, the functions $(1 - 2\pi Z^2)e^{-\pi Z^2}$ and $2\pi Z^2 e^{-\pi Z^2}$ have moments $(1, 0)$ and $(0, 1)$. They split H into $H_0 \oplus \mathbb{C}^2$, so $M_0 = (R \otimes H_0) \oplus (I \otimes \mathbb{C}^2)$. Dividing by IM_0 gives

$$Q_0 = H_0 \oplus (I/I^2) \otimes \mathbb{C}^2, \quad I/I^2 \cong \bigoplus_p \mathbb{C}\ell_p.$$

The last isomorphism follows from $t_{ab} - 1 = (t_a - 1) + (t_b - 1) + (t_a - 1)(t_b - 1)$ and prime factorization; the valuations v_p kill I^2 and separate the displayed generators. This proves ER29 and the claimed detection without a new quotient assumption.

For every prime p , the particular class is now evaluated:

$$b_{p,t} = [(t_p - 1) \otimes h_t], \quad \boxed{\beta b_{p,t} = \ell_p \otimes \left(H_t(0), \frac{\sqrt{\pi t}}{8} e^{-1/(4t)} \right) \neq 0.} \quad (\text{ER30})$$

Its original sum-of-components map is zero. Therefore the original periodization $\mathcal{E}(c)(x) = 2x^{1/2} \sum_{n \geq 1} (\Phi c)(nx)$ and its full $\int_0^\infty |\mathcal{E}(c)(x)|^2 (1 + \log^2 x)^{\delta/2} dx/x$, $\delta > 1$, vanish on this class. These are the moment and periodization conventions of [Alain Connes, *Trace formula in non-commutative geometry and the zeros of the Riemann zeta function*](#), Section III (6)–(19). The algebraic prime quotient and its proved source comparison are [FR15–25](#). No new positive norm on these zero-seminorm classes is assigned.

There is more information than these two moments. Put $w = s - \frac{1}{2}$ and retain the full Mellin convention $\mathcal{M}h(s) = \int_{\mathbb{R}} h(Z) e^{-wZ} dZ$. For every complex s , the complete test is

$$\mathcal{M}h_t(s) = \sqrt{\pi t} \left\{ e^{t(w-i/(2t))^2} H_0(2tw - i) + e^{t(w+i/(2t))^2} H_0(2tw + i) \right\}. \quad (\text{ER31})$$

To prove this, first omit the cosine in ER27. Gaussian integration with the complete H_t source gives

$$\int_{\mathbb{R}} e^{-Z^2/(4t)} H_t(Z) e^{-wZ} dZ = 2\sqrt{\pi t} e^{tw^2} H_0(2tw). \quad (\text{ER32})$$

Absolute interchange holds for every fixed complex w , since the outer Gaussian dominates $e^{|\Re w||Z|}$, and $\int e^{tu^2} |\Phi(u)| du < \infty$. The inner Gaussian formula is entire in w . The two exponentials of the retained cosine give ER31.

In terms of the original zeta function, set $\sigma_0 = it(s - \frac{1}{2})$ and $\sigma_1 = 1 + it(s - \frac{1}{2})$. ER3 gives the exact formula

$$\boxed{\mathcal{M}h_t(s) = \frac{\sqrt{\pi t}}{16} \left\{ e^{t(w-i/(2t))^2} \sigma_1(\sigma_1 - 1) \pi^{-\sigma_1/2} \Gamma(\sigma_1/2) \zeta(\sigma_1) \right.} \\ \left. + e^{t(w+i/(2t))^2} \sigma_0(\sigma_0 - 1) \pi^{-\sigma_0/2} \Gamma(\sigma_0/2) \zeta(\sigma_0) \right\}.} \quad (\text{ER33})$$

All endpoint factors, Gamma factors and powers of pi are shown. Each product is continued as the product in ER3 at every zero or pole of a factor. At $s = 1/2$, the two exceptional arguments are exactly 1 and 0. Their products are both 1: the first is ER4, and the second follows from $C(0) = -2$, $\zeta(0) = -1/2$. Substitution in ER33 reproduces r_t in ER28. Both endpoints thus contribute to the evaluated prime-boundary coordinate. The endpoint values $\mathcal{M}h_t(0)$ and $\mathcal{M}h_t(1)$ are also the full expressions ER33 at $w = -1/2$ and $w = 1/2$; no endpoint-null test has been imposed.

1.5 An invertible test filter and the complete arithmetic pairing

The just-constructed source can be taken to an endpoint-null test without losing its information. This requires the inverse, not merely the assertion that the endpoint values vanish. In the same real coordinate put

$$f_t = (\partial_Z^2 - \tfrac{1}{4})h_t, \quad F_t(s) = s(s-1)\mathcal{M}h_t(s), \quad k_t = f_t * f_t. \quad (\text{ER34})$$

Gaussian decay of h_t and every derivative follows from ER27, because all derivatives of H_t are bounded on the real axis. Thus integration by parts proves $\mathcal{M}f_t = F_t$, with the two endpoints $F_t(0) = F_t(1) = 0$ evaluated exactly. ER33 remains the full formula for $\mathcal{M}h_t$; its factors are not omitted.

The distribution $G(Z) = -e^{-|Z|/2}$ satisfies $(\partial_Z^2 - \frac{1}{4})G = \delta_0$: the derivative jump of $e^{-|Z|/2}$ is -1 , fixing the sign and coefficient. The inverse on the actual Schwartz source is

$$h_t(Z) = - \int_{\mathbb{R}} e^{-|Z-y|/2} f_t(y) dy. \quad (\text{ER35})$$

More generally convolution with G is continuous on Schwartz space. Move derivatives to the Schwartz factor and use $(1 + |Z|)^M \leq (1 + |Z - y|)^M (1 + |y|)^M$; every polynomially weighted exponential is integrable. The only solutions of the homogeneous differential equation are $ae^{Z/2} + be^{-Z/2}$, none nonzero and Schwartz. This proves the two-sided inverse. Although the displayed convolution permits exponential tails for a generic Schwartz input, for the actual f_t it returns exactly the Gaussian-decaying h_t ; the endpoint equalities are retained in that assertion.

The prime-boundary coordinates of this filtered source are also fully evaluated:

$$\beta[(t_p - 1) \otimes f_t] = \ell_p \otimes \left(H_t''(0) - \left(\frac{1}{2t} + \frac{1}{4t^2} + \frac{1}{4} \right) H_t(0), -\frac{\sqrt{\pi t}}{32} e^{-1/(4t)} \right). \quad (\text{ER36})$$

Differentiate ER27 twice at zero to obtain its first entry; integrate f_t and use $\int h_t'' = 0$ for the second. Every summand in $\Phi(u)$ is positive on $u \geq 0$, since $2\pi n^2 e^{4u} > 3$. Hence

$$H_t(0) > 0, \quad H_t''(0) = - \int_0^\infty u^2 e^{tu^2} \Phi(u) du < 0,$$

and both entries of ER36 are strictly negative. These are signs of two actual source moments. They are not asserted to be signs of the Weil quadratic form.

For completeness we give that quadratic form's exact receiving formula, with all original arithmetic contributions. Let

$$\begin{aligned} j(s) &= \zeta'(s)/\zeta(s), \\ \kappa(s) &= -\frac{1}{2} \log \pi + \frac{1}{2} \psi_\Gamma(s/2), \\ A_t(s) &= F_t(s)^2, \quad d_{m,t} = A_t(-2m), \quad U_{N,t} = \sum_{m=1}^N d_{m,t}, \\ K_t &= \sum_{\rho} m_{\rho} F_t(\rho)^2, \quad V_{N,t} = K_t + U_{N,t} - A_t(1). \end{aligned} \quad (\text{ER37})$$

The sum in K_t is over the actual nontrivial zeros with multiplicity. On the present real even source the reflected test is the same F_t : indeed $\overline{F_t(1 - \bar{s})} = F_t(s)$. This proves its square in ER37 rather than replacing it by an absolute square away from the critical line.

The Gaussian bounds for f_t and its derivatives imply Gaussian bounds for k_t and its derivatives by convolution and $y^2 + (x - y)^2 = x^2/2 + 2(y - x/2)^2$. For example fixed polynomials times $e^{-Z^2/(4t)}$ bound each derivative of f_t ; powers can be bounded by constants times $e^{-Z^2/(8t)}$. Integration by parts in $\mathcal{M}f_t$ then gives arbitrary inverse-power decay on every fixed vertical strip. The original zero count $O(T \log(T + 2))$ proves absolute convergence of the nontrivial-zero sum. These are the same zero-count and contour hypotheses used in the classical explicit formula. For the human source conventions see Connes, cited above, Appendix II (1)–(11), Theorem 1 and Theorem 6. The latter theorem is stated for compact support. The present Gaussian source is noncompact; the displayed Gaussian bounds and the finite-strip argument below establish the needed extension, rather than silently treating it as a compactly supported input. The original reflection identity, obtained by logarithmically differentiating ER3 and its reflection while retaining both Gamma factors, is

$$j(s) + j(1 - s) = -\kappa(s) - \kappa(1 - s). \quad (\text{ER38})$$

Let I_c denote the upward vertical integral divided by $2\pi i$, take $\sigma > 1$, and $b_N = -2N - 1$. The original argument principle on this finite-width strip gives

$$\begin{aligned}
V_{N,t} &= I_\sigma(A_t j) - I_{b_N}(A_t j), \\
\mathcal{G}_{N,t} &= I_{b_N}\{A_t(s)[\kappa(s) + \kappa(1-s)]\}, \\
V_{N,t} &= -P_t + \mathcal{G}_{N,t}, \\
\mathcal{G}_{N,t} &= A_\infty(k_t) + A_t(0) + U_{N,t}, \\
K_t &= A_t(0) + A_t(1) + A_\infty(k_t) - P_t.
\end{aligned} \tag{ER39}$$

Here the complete terms are

$$\begin{aligned}
P_t &= 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} k_t(\log n), \quad \Lambda(p^j) = \log p, \\
A_\infty(k) &= -(\gamma_E + \log \pi)k(0) \\
&\quad + \int_0^\infty \frac{e^{-x}k(0) - \frac{1}{2}e^{-x/4}[k(x/2) + k(-x/2)]}{1 - e^{-x}} dx, \\
A_t(0) &= A_t(1) = 0.
\end{aligned} \tag{ER40}$$

Both prime legs occur in the factor two because k_t is even. The Gamma numerator is kept together: its first-order difference is $O(x)$ and its symmetric second difference is $O(x^2)$ at zero. The Gaussian tails prove absolute convergence of the prime sum, including all prime powers, and of the remaining Gamma integral.

To verify ER39, exhaust the strip in height along levels avoiding zero ordinates. The logarithmic derivative of the original function, with the full derivative of C retained in its factored expansion, has polynomial bounds along such a sequence; the vertical test's arbitrary inverse-power decay kills the horizontal edges. The enclosed original divisor is exactly the nontrivial zeros, the first N negative even integers, and the pole at one. Inserting ER38 on the left edge and using $j(s) = -\sum_{n \geq 2} \Lambda(n)n^{-s}$ on the right gives the two prime legs by Mellin inversion. Shifting the Gamma edge to $1/2$ crosses residues -1 at $0, -2, \dots, -2N$, and therefore adds precisely $A_t(0) + U_{N,t}$ to its value on the left. The line $1/2$ gives A_∞ by the original digamma integral, whose full numerator is ER40. This proves all signs and every finite-cutoff term. No limit of $U_{N,t}$ is presupposed or assigned by this calculation.

The full input to ER39 has thus been constructed: F_t is the two-argument original-zeta expression ER33 multiplied by the retained $s(s-1)$, and k_t is its actual source convolution. The map ER35 recovers the source whose prime-boundary residue was evaluated in ER30. Formula ER39 gives its exact compensated pairing while retaining the original signed trace, each trivial zero and the matching Gamma boundary separately. This extension does not claim a numerical value or sign for K_t .

The full support carrier is retained. For any of the displayed complex vector spaces V and the original bounded distributive lattice L , retain the carrier

$$G_L(V) = \{(v, 1_L) : v \in V\} \cup \{z_\lambda = (0, \lambda) : \lambda \neq 1_L\}, \quad e = (0, 1_L), \quad \tau = z_{0_L}. \tag{ER40a}$$

Each linear map $A : V \rightarrow W$ proved above has the specified lift $(v, 1_L) \mapsto (Av, 1_L)$, $z_\lambda \mapsto z_\lambda$. Composition of these lifts agrees with the lift of the composition, and every original label is unchanged even when the amplitude vanishes. The injective enlarged-source map ER24 remains injective on these carriers: this follows on top amplitudes from ER24 and on the distinct lower points from their unchanged labels. The two different lateral inputs in ER26 have the same top-supported image; the difference of their images has amplitude zero with label 1_L , namely e , rather than the absent label τ . This construction uses the full L , without replacing it

by a Boolean quotient. It is a lift of the proved linear source maps; it does not assert that vector-space differential operators are semiring homomorphisms, or replace the separately proved supported-zero multiplication by an identity operator.

ER11, ER21, ER24–26 and ER30–40a give the exact relation sought: the original arithmetic residue obstructs a tempered unfiltered positive-time source, the two actual endpoint continuations provide a faithful source beyond that domain, and an explicit Schwartz receiver carries the obstruction value into the existing arithmetic prime class and a two-argument original-zeta test. The nonzero residue occurs independently of RH. This calculation does not assign the sign of a compensated Weil form or locate a nontrivial zero; those require their own evaluated arithmetic pairings.